

Class 12 Physics

Complete Formula Handbook

Rapid Revision • All Important Formulas • Large Font Edition

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1 Electric Charges and Fields

Basic Electrostatics & Coulomb's Law

- **Quantization of Charge:**

$$q = \pm ne \quad (e = 1.6 \times 10^{-19} \text{ C})$$

- **Coulomb's Law (Scalar):**

$$F = \frac{1}{4\pi\epsilon_0} \frac{|q_1 q_2|}{r^2}, \quad \frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2$$

- **Coulomb's Law in Medium:**

$$F_m = \frac{F_0}{\epsilon_r} = \frac{F_0}{K} \quad (\epsilon = \epsilon_0 K)$$

- **Coulomb's Law (Vector Form):**

$$\vec{F}_{12} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{|\vec{r}_1 - \vec{r}_2|^3} (\vec{r}_1 - \vec{r}_2)$$

Electric Field & Electric Dipole

- **Electric Field due to Point Charge:**

$$\vec{E} = \frac{\vec{F}}{q_0} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{r}$$

- **Electric Dipole Moment:**

$$\vec{p} = q(2\vec{a}) \quad (\text{directed from } -q \text{ to } +q)$$

- **Axial Electric Field (Short Dipole, $r \gg a$):**

$$\vec{E}_{\text{axial}} = \frac{1}{4\pi\epsilon_0} \frac{2\vec{p}}{r^3}$$

- **Equatorial Electric Field (Short Dipole, $r \gg a$):**

$$\vec{E}_{\text{eq}} = -\frac{1}{4\pi\epsilon_0} \frac{\vec{p}}{r^3}$$

- **General Point (r, θ) Field:**

$$E = \frac{1}{4\pi\epsilon_0} \frac{p}{r^3} \sqrt{1 + 3\cos^2\theta}, \quad \tan\alpha = \frac{1}{2} \tan\theta$$

- **Torque on Dipole:**

$$\vec{\tau} = \vec{p} \times \vec{E}, \quad \tau = pE \sin\theta$$

- **Potential Energy of Dipole:**

$$U = -\vec{p} \cdot \vec{E} = -pE \cos\theta$$

- **Work Done in Rotating Dipole from θ_1 to θ_2 :**

$$W = pE(\cos\theta_1 - \cos\theta_2)$$

Gauss's Law and Its Applications

- **Electric Flux:**

$$\Phi_E = \int \vec{E} \cdot d\vec{A} = EA \cos \theta$$

- **Gauss's Law:**

$$\Phi_{\text{net}} = \oint \vec{E} \cdot d\vec{A} = \frac{q_{\text{enclosed}}}{\epsilon_0}$$

- **Field due to Infinitely Long Straight Wire:**

$$E = \frac{\lambda}{2\pi\epsilon_0 r}$$

- **Field due to Infinite Plane Sheet:**

$$E = \frac{\sigma}{2\epsilon_0}$$

- **Field between Two Oppositely Charged Plates:**

$$E_{\text{inside}} = \frac{\sigma}{\epsilon_0}, \quad E_{\text{outside}} = 0$$

- **Field due to Thin Spherical Shell (Radius R):**

$$E_{\text{out}} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \quad (r > R), \quad E_{\text{surface}} = \frac{1}{4\pi\epsilon_0} \frac{q}{R^2}, \quad E_{\text{in}} = 0 \quad (r < R)$$

2 Electrostatic Potential and Capacitance

Electric Potential & Potential Energy

- **Potential due to a Point Charge:**

$$V = \frac{1}{4\pi\epsilon_0} \frac{q}{r}$$

- **Potential due to a Dipole at (r, θ) :**

$$V = \frac{1}{4\pi\epsilon_0} \frac{p \cos \theta}{r^2} \quad \left(V_{\text{axial}} = \frac{p}{4\pi\epsilon_0 r^2}, \quad V_{\text{eq}} = 0 \right)$$

- **Relation between Field and Potential:**

$$\vec{E} = -\vec{\nabla}V \implies E = -\frac{dV}{dr}$$

- **Potential Energy of Two Point Charges:**

$$U = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r_{12}}$$

- **Potential Energy in External Field $\vec{E}$:**

$$U = q_1 V(\vec{r}_1) + q_2 V(\vec{r}_2) + \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r_{12}}$$

Capacitance and Dielectrics

- **Definition of Capacitance:**

$$C = \frac{Q}{V}$$

- **Isolated Spherical Conductor (Radius R):**

$$C = 4\pi\epsilon_0 R$$

- **Parallel Plate Capacitor (Vacuum):**

$$C_0 = \frac{\epsilon_0 A}{d}$$

- **Parallel Plate Capacitor with Full Dielectric (K):**

$$C = KC_0 = \frac{K\epsilon_0 A}{d}$$

- **Capacitor with Dielectric Slab of Thickness $t < d$:**

$$C = \frac{\epsilon_0 A}{d - t\left(1 - \frac{1}{K}\right)}$$

- **Capacitor with Conducting Slab of Thickness $t < d$:**

$$C = \frac{\epsilon_0 A}{d - t}$$

Combinations of Capacitors & Energy Stored

- **Capacitors in Series:**

$$\frac{1}{C_s} = \frac{1}{C_1} + \frac{1}{C_2} + \cdots + \frac{1}{C_n} \quad (\text{Charge } Q \text{ is constant})$$

- **Capacitors in Parallel:**

$$C_p = C_1 + C_2 + \cdots + C_n \quad (\text{Voltage } V \text{ is constant})$$

- **Electrostatic Energy Stored in Capacitor:**

$$U = \frac{1}{2}CV^2 = \frac{Q^2}{2C} = \frac{1}{2}QV$$

- **Electrostatic Energy Density:**

$$u = \frac{1}{2}\varepsilon_0 E^2 \quad \left(\text{in medium: } u = \frac{1}{2}K\varepsilon_0 E^2 \right)$$

- **Common Potential on Connecting Two Capacitors:**

$$V_{\text{common}} = \frac{C_1 V_1 + C_2 V_2}{C_1 + C_2}$$

- **Loss of Energy on Sharing Charges:**

$$\Delta U = \frac{C_1 C_2 (V_1 - V_2)^2}{2(C_1 + C_2)}$$

3 Current Electricity

Electric Current & Conduction Mechanisms

- **Electric Current:**

$$I = \frac{dQ}{dt}, \quad I_{\text{avg}} = \frac{Q}{t} = \frac{ne}{t}$$

- **Drift Velocity:**

$$v_d = \frac{eE\tau}{m} = \frac{eV\tau}{ml}$$

- **Current and Drift Velocity Relation:**

$$I = neAv_d$$

- **Current Density:**

$$j = \frac{I}{A} = nev_d = \sigma E$$

- **Electron Mobility:**

$$\mu = \frac{v_d}{E} = \frac{e\tau}{m} \implies I = neA\mu E$$

- **Resistivity and Conductivity:**

$$\rho = \frac{m}{ne^2\tau}, \quad \sigma = \frac{1}{\rho} = \frac{ne^2\tau}{m}$$

- **Resistance Formula:**

$$R = \rho \frac{l}{A} = \frac{ml}{ne^2\tau A}$$

- **Temperature Dependence of Resistance:**

$$R_T = R_0(1 + \alpha\Delta T), \quad \rho_T = \rho_0(1 + \alpha\Delta T)$$

Combinations of Resistors, EMF & Cells

- **Resistors in Series:**

$$R_s = R_1 + R_2 + \cdots + R_n$$

- **Resistors in Parallel:**

$$\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} + \cdots + \frac{1}{R_n}$$

- **Terminal Potential Difference & EMF:**

$$V = \mathcal{E} - Ir \quad (\text{Discharging}), \quad V = \mathcal{E} + Ir \quad (\text{Charging})$$

- **Internal Resistance:**

$$r = \left(\frac{\mathcal{E}}{V} - 1 \right) R$$

- **Cells in Series:**

$$\mathcal{E}_{\text{eq}} = \mathcal{E}_1 + \mathcal{E}_2, \quad r_{\text{eq}} = r_1 + r_2$$

- **Cells in Parallel:**

$$\mathcal{E}_{\text{eq}} = \frac{\mathcal{E}_1 r_2 + \mathcal{E}_2 r_1}{r_1 + r_2}, \quad r_{\text{eq}} = \frac{r_1 r_2}{r_1 + r_2} \quad \left(\frac{\mathcal{E}_{\text{eq}}}{r_{\text{eq}}} = \sum \frac{\mathcal{E}_i}{r_i} \right)$$

- **Maximum Power Transfer Theorem:**

$$P_{\text{max}} = \frac{\mathcal{E}^2}{4r} \quad (\text{when external resistance } R = r)$$

Kirchhoff's Laws & Measuring Instruments

- **Kirchhoff's Current Law (Junction Rule):**

$$\sum I = 0 \quad (\text{Law of Conservation of Charge})$$

- **Kirchhoff's Voltage Law (Loop Rule):**

$$\sum \Delta V = 0 \quad (\text{Law of Conservation of Energy})$$

- **Wheatstone Bridge Principle (Balanced Condition):**

$$\frac{P}{Q} = \frac{R}{S} \quad (\text{Galvanometer current } I_g = 0)$$

- **Meter Bridge (Unknown Resistance):**

$$S = \left(\frac{100 - l}{l} \right) R$$

- **Electric Power & Joule's Heating:**

$$P = VI = I^2R = \frac{V^2}{R}, \quad H = I^2Rt$$

4 Moving Charges and Magnetism

Lorentz Force & Biot-Savart Law

- **Lorentz Force:**

$$\vec{F} = q(\vec{E} + \vec{v} \times \vec{B})$$

- **Magnetic Force on Moving Charge:**

$$F_m = qvB \sin \theta \quad (\text{Zero work done by magnetic force: } W = 0)$$

- **Radius & Time Period in Circular Orbit ($\vec{v} \perp \vec{B}$):**

$$r = \frac{mv}{qB} = \frac{p}{qB} = \frac{\sqrt{2mK}}{qB}, \quad T = \frac{2\pi m}{qB}, \quad \nu = \frac{qB}{2\pi m}$$

- **Pitch of Helical Path ($\theta \neq 90^\circ$):**

$$p = v_{\parallel} T = (v \cos \theta) \frac{2\pi m}{qB}$$

- **Velocity Selector Condition:**

$$v = \frac{E}{B}$$

- **Biot-Savart Law:**

$$d\vec{B} = \frac{\mu_0}{4\pi} \frac{I(d\vec{l} \times \hat{r})}{r^2}, \quad \frac{\mu_0}{4\pi} = 10^{-7} \text{ T} \cdot \text{m/A}$$

Magnetic Field Formulas

- **Center of Circular Loop of N Turns:**

$$B_{\text{center}} = \frac{\mu_0 N I}{2R}$$

- **Axis of Circular Current Loop (Distance x):**

$$B_{\text{axis}} = \frac{\mu_0 N I R^2}{2(R^2 + x^2)^{3/2}}$$

- **Straight Wire of Finite Length:**

$$B = \frac{\mu_0 I}{4\pi d} (\sin \phi_1 + \sin \phi_2)$$

- **Infinitely Long Straight Conductor:**

$$B = \frac{\mu_0 I}{2\pi d}$$

- **Ampere's Circuital Law:**

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{\text{enclosed}}$$

- **Ideal Solenoid:**

$$B = \mu_0 n I = \frac{\mu_0 N I}{L} \quad \left(B_{\text{end}} = \frac{1}{2} \mu_0 n I \right)$$

- **Toroid (Radius r):**

$$B = \mu_0 n I = \frac{\mu_0 N I}{2\pi r}$$

Force on Conductors & Galvanometer

- **Force on Current-Carrying Conductor:**

$$\vec{F} = I(\vec{L} \times \vec{B}) \implies F = ILB \sin \theta$$

- **Force Between Two Parallel Current Wires (Separation r):**

$$\frac{F}{L} = \frac{\mu_0 I_1 I_2}{2\pi r} \quad (\text{Same dir: Attract; Opposite dir: Repel})$$

- **Definition of 1 Ampere:** Force of 2×10^{-7} N/m per unit length in vacuum.
- **Magnetic Dipole Moment of Loop:**

$$\vec{M} = NI\vec{A}$$

- **Torque on Current Loop in Magnetic Field:**

$$\vec{\tau} = \vec{M} \times \vec{B} \implies \tau = NIAB \sin \theta = MB \sin \theta$$

- **Moving Coil Galvanometer Equation:**

$$I = \left(\frac{C}{NAB} \right) \theta = K\theta$$

- **Current Sensitivity & Voltage Sensitivity:**

$$I_s = \frac{\theta}{I} = \frac{NAB}{C}, \quad V_s = \frac{\theta}{V} = \frac{NAB}{CR}$$

- **Galvanometer to Ammeter Shunt:**

$$S = \frac{I_g G}{I - I_g}$$

- **Galvanometer to Voltmeter Series Resistance:**

$$R = \frac{V}{I_g} - G$$

5 Magnetism and Matter

Bar Magnet and Terrestrial Magnetism

- **Bar Magnet Magnetic Moment:**

$$M = m(2l) \quad (m = \text{pole strength})$$

- **Axial Field of Short Magnet ($d \gg l$):**

$$B_{\text{axial}} = \frac{\mu_0}{4\pi} \frac{2M}{d^3}$$

- **Equatorial Field of Short Magnet ($d \gg l$):**

$$B_{\text{eq}} = \frac{\mu_0}{4\pi} \frac{M}{d^3} \quad (B_{\text{axial}} = 2B_{\text{eq}})$$

- **Torque & Potential Energy of Bar Magnet:**

$$\vec{\tau} = \vec{M} \times \vec{B}, \quad U = -\vec{M} \cdot \vec{B} = -MB \cos \theta$$

- **Time Period of Magnetic Dipole in Magnetic Field:**

$$T = 2\pi \sqrt{\frac{I_{\text{moment}}}{MB}}$$

- **Earth's Magnetic Field Components:**

$$B_H = B_E \cos \delta, \quad B_V = B_E \sin \delta, \quad \tan \delta = \frac{B_V}{B_H}, \quad B_E = \sqrt{B_H^2 + B_V^2}$$

- **Gauss's Law in Magnetism:**

$$\oint \vec{B} \cdot d\vec{A} = 0 \quad (\text{Magnetic monopoles do not exist})$$

Magnetic Properties of Materials

- **Magnetization & Magnetic Intensity:**

$$\vec{M} = \frac{\vec{m}_{\text{net}}}{\text{Volume}}, \quad \vec{B} = \mu_0(\vec{H} + \vec{M})$$

- **Magnetic Susceptibility & Relative Permeability:**

$$\chi_m = \frac{M}{H}, \quad \mu_r = 1 + \chi_m, \quad \mu = \mu_0\mu_r = \mu_0(1 + \chi_m)$$

- **Curie's Law (Paramagnetic Materials):**

$$\chi_m = \frac{C}{T} \quad (T = \text{Absolute Temperature})$$

- **Curie-Weiss Law (Ferromagnetic above T_c):**

$$\chi_m = \frac{C}{T - T_c} \quad (T > T_c)$$

6 Electromagnetic Induction

Magnetic Flux & Faraday-Lenz Laws

- **Magnetic Flux:**

$$\Phi_B = \vec{B} \cdot \vec{A} = BA \cos \theta \quad (\text{SI Unit: Weber, Wb})$$

- **Faraday's Law of Induction:**

$$\mathcal{E} = -\frac{d\Phi_B}{dt} = -N\frac{d\Phi_B}{dt}$$

- **Induced Current and Charge Transferred:**

$$I = \frac{|\mathcal{E}|}{R} = \frac{1}{R} \frac{d\Phi_B}{dt}, \quad \Delta Q = \frac{\Delta\Phi_B}{R} \quad (\text{Charge is time-independent})$$

- **Motional EMF (Translational):**

$$\mathcal{E} = Blv \sin \theta \quad (\vec{v} \perp \vec{B} \perp \vec{l} \implies \mathcal{E} = Blv)$$

- **Motional EMF (Rotating Rod about one end):**

$$\mathcal{E} = \frac{1}{2} B \omega l^2 = B \nu \pi l^2$$

- **External Force and Power Required to Move Rod:**

$$F = \frac{B^2 l^2 v}{R}, \quad P = Fv = \frac{B^2 l^2 v^2}{R}$$

Inductance & AC Generator

- **Self-Inductance Definition:**

$$\Phi = LI \implies \mathcal{E} = -L \frac{dI}{dt}$$

- **Self-Inductance of a Long Solenoid:**

$$L = \frac{\mu_0 N^2 A}{l} = \mu_0 n^2 Al \quad (\text{with core } \mu_r : L = \mu_0 \mu_r n^2 Al)$$

- **Magnetic Potential Energy Stored in Inductor:**

$$U = \frac{1}{2} LI^2$$

- **Magnetic Energy Density:**

$$u_B = \frac{B^2}{2\mu_0}$$

- **Mutual Inductance Definition:**

$$\Phi_2 = MI_1 \implies \mathcal{E}_2 = -M \frac{dI_1}{dt}$$

- **Mutual Inductance of Coaxial Solenoids:**

$$M = \frac{\mu_0 N_1 N_2 A}{l} = \mu_0 n_1 n_2 Al$$

- **Coupling Coefficient:**

$$M = k\sqrt{L_1 L_2} \quad (0 \leq k \leq 1)$$

- **AC Generator Output EMF:**

$$\mathcal{E} = NBA\omega \sin \omega t = \mathcal{E}_0 \sin \omega t \quad (\mathcal{E}_0 = NBA\omega)$$

7 Alternating Current

AC Fundamentals

- **Alternating EMF & Current:**

$$V = V_0 \sin \omega t, \quad I = I_0 \sin(\omega t \pm \phi)$$

- **Mean / Average Values (Half Cycle):**

$$V_{\text{avg}} = \frac{2}{\pi} V_0 \approx 0.637 V_0, \quad I_{\text{avg}} = \frac{2}{\pi} I_0 \approx 0.637 I_0$$

(Average value over a complete full cycle is zero: $V_{\text{avg, full}} = 0$)

- **RMS / Effective Values:**

$$V_{\text{rms}} = \frac{V_0}{\sqrt{2}} \approx 0.707 V_0, \quad I_{\text{rms}} = \frac{I_0}{\sqrt{2}} \approx 0.707 I_0$$

Reactance and Impedance of AC Circuits

- **Pure Resistor (R):**

$$V \text{ and } I \text{ in phase, } \phi = 0, \quad I_0 = \frac{V_0}{R}$$

- **Pure Inductor (L):**

$$I \text{ lags } V \text{ by } \frac{\pi}{2}, \quad X_L = \omega L = 2\pi f L, \quad I_0 = \frac{V_0}{X_L}$$

- **Pure Capacitor (C):**

$$I \text{ leads } V \text{ by } \frac{\pi}{2}, \quad X_C = \frac{1}{\omega C} = \frac{1}{2\pi f C}, \quad I_0 = \frac{V_0}{X_C}$$

- **Series LCR Circuit Impedance:**

$$Z = \sqrt{R^2 + (X_L - X_C)^2} = \sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}$$

- **Phase Angle:**

$$\tan \phi = \frac{X_L - X_C}{R} = \frac{\omega L - \frac{1}{\omega C}}{R}$$

- **Series Resonance Condition ($X_L = X_C$):**

$$\omega_r = \frac{1}{\sqrt{LC}}, \quad f_r = \frac{1}{2\pi\sqrt{LC}}, \quad Z_{\min} = R, \quad I_{\max} = \frac{V_0}{R}$$

- **Quality Factor (Q -Factor):**

$$Q = \frac{\omega_r L}{R} = \frac{1}{\omega_r C R} = \frac{1}{R} \sqrt{\frac{L}{C}} = \frac{\omega_r}{2\Delta\omega} = \frac{f_r}{\text{Bandwidth}}$$

Power in AC Circuits & Transformer

- **Average AC Power:**

$$P_{\text{avg}} = V_{\text{rms}} I_{\text{rms}} \cos \phi$$

- **Power Factor:**

$$\cos \phi = \frac{R}{Z} \quad (\text{Pure } R : \cos \phi = 1; \text{ Pure } L \text{ or } C : \cos \phi = 0, P = 0)$$

- **Wattless Current (Idle Current):**

$$I_{\text{wattless}} = I_{\text{rms}} \sin \phi$$

- **Transformer Transformation Ratio (K):**

$$\frac{V_s}{V_p} = \frac{N_s}{N_p} = \frac{I_p}{I_s} = K$$

- **Transformer Efficiency:**

$$\eta = \frac{\text{Output Power}}{\text{Input Power}} \times 100\% = \frac{V_s I_s}{V_p I_p} \times 100\%$$

8 Electromagnetic Waves

Displacement Current & Maxwell's Equations

- Displacement Current:

$$I_d = \varepsilon_0 \frac{d\Phi_E}{dt}$$

- Ampere-Maxwell Law:

$$\oint \vec{B} \cdot d\vec{l} = \mu_0(I_c + I_d) = \mu_0 I_c + \mu_0 \varepsilon_0 \frac{d\Phi_E}{dt}$$

- Speed of Electromagnetic Wave in Vacuum:

$$c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}} = 3 \times 10^8 \text{ m/s}$$

- Speed of EM Wave in Material Medium:

$$v = \frac{1}{\sqrt{\mu \varepsilon}} = \frac{c}{\sqrt{\mu_r \varepsilon_r}} = \frac{c}{n}$$

- Relation between Electric and Magnetic Amplitudes:

$$\frac{E_0}{B_0} = \frac{E}{B} = c$$

- Electromagnetic Energy Density:

$$u = u_E + u_B = \frac{1}{2} \varepsilon_0 E^2 + \frac{B^2}{2\mu_0} = \varepsilon_0 E_{\text{rms}}^2$$

- Intensity of EM Wave:

$$I = u_{\text{avg}} c = \frac{1}{2} \varepsilon_0 E_0^2 c = \varepsilon_0 E_{\text{rms}}^2 c$$

- Radiation Pressure (Complete Absorption vs Reflection):

$$P_{\text{abs}} = \frac{I}{c}, \quad P_{\text{refl}} = \frac{2I}{c}$$

Electromagnetic Spectrum (Decreasing λ , Increasing ν)

Wave Type	Wavelength (λ) Range	Primary Use
Radio waves	$> 0.1 \text{ m}$	Radio & TV communications
Microwaves	$0.1 \text{ m} - 1 \text{ mm}$	RADAR, Microwave ovens
Infrared	$1 \text{ mm} - 700 \text{ nm}$	Night vision, Remote controls
Visible light	$700 \text{ nm} - 400 \text{ nm}$	Human vision
Ultraviolet	$400 \text{ nm} - 1 \text{ nm}$	Water sterilization, LASIK
X-rays	$1 \text{ nm} - 10^{-3} \text{ nm}$	Medical imaging, Crystallography
Gamma rays	$< 10^{-3} \text{ nm}$	Cancer radiotherapy, Nuclear study

9 Ray Optics and Optical Instruments

Reflection and Refraction Formulas

- **Mirror Formula and Focal Length:**

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}, \quad f = \frac{R}{2}$$

- **Linear Magnification for Mirrors:**

$$m = \frac{h_i}{h_o} = -\frac{v}{u} = \frac{f - v}{f} = \frac{f}{f - u}$$

- **Snell's Law of Refraction:**

$$n_{21} = \frac{n_2}{n_1} = \frac{\sin i}{\sin r} = \frac{v_1}{v_2} = \frac{\lambda_1}{\lambda_2}$$

- **Real Depth and Apparent Depth:**

$$\text{Apparent Depth} = \frac{\text{Real Depth}}{n}, \quad \text{Normal Shift} = t \left(1 - \frac{1}{n} \right)$$

- **Critical Angle and Total Internal Reflection (TIR):**

$$\sin C = \frac{1}{n} = \frac{n_{\text{rarer}}}{n_{\text{denser}}}$$

- **Refraction at a Single Spherical Surface:**

$$\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$$

Lenses, Prisms & Optical Instruments

- **Lens Maker's Formula:**

$$\frac{1}{f} = \left(\frac{n_2}{n_1} - 1 \right) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

- **Thin Lens Formula:**

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}, \quad m = \frac{h_i}{h_o} = \frac{v}{u}$$

- **Power of a Lens:**

$$P = \frac{1}{f \text{ (in meters)}} = \frac{100}{f \text{ (in cm)}} \quad (\text{Diopter, D})$$

- **Combination of Thin Lenses in Contact:**

$$\frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2} + \dots, \quad P = P_1 + P_2 + \dots, \quad m = m_1 \times m_2 \times \dots$$

- **Prism Formula:**

$$A = r_1 + r_2, \quad \delta = i + e - A, \quad n = \frac{\sin\left(\frac{A+D_m}{2}\right)}{\sin\left(\frac{A}{2}\right)}$$

- **Small Angle Prism Deviation:**

$$\delta = (n - 1)A$$

- **Simple Microscope Magnification:**

$$m = 1 + \frac{D}{f} \quad (\text{at Near Point } D = 25 \text{ cm}), \quad m = \frac{D}{f} \quad (\text{at Infinity})$$

- **Compound Microscope Magnification:**

$$m = m_o \times m_e = \left(\frac{v_o}{u_o} \right) \left(1 + \frac{D}{f_e} \right), \quad m_\infty \approx \frac{L}{f_o f_e}$$

- **Astronomical Telescope Magnification:**

$$m = \frac{f_o}{f_e} \left(1 + \frac{f_e}{D} \right) \quad (\text{Near Point}), \quad m = \frac{f_o}{f_e} \quad (\text{Normal adjustment}), \quad L = f_o + f_e$$

10 Wave Optics

Huygens' Principle & Interference (YDSE)

- **Resultant Amplitude and Intensity of Two Waves:**

$$I = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos \phi$$

- **For Identical Sources ($I_1 = I_2 = I_0$):**

$$I = 4I_0 \cos^2 \left(\frac{\phi}{2} \right)$$

- **Maximum and Minimum Intensities:**

$$I_{\max} = (\sqrt{I_1} + \sqrt{I_2})^2, \quad I_{\min} = (\sqrt{I_1} - \sqrt{I_2})^2$$

$$\frac{I_{\max}}{I_{\min}} = \left(\frac{A_1 + A_2}{A_1 - A_2} \right)^2 = \left(\frac{r + 1}{r - 1} \right)^2 \quad \left(r = \frac{A_1}{A_2} \right)$$

- **Path Difference and Phase Difference:**

$$\Delta \phi = \frac{2\pi}{\lambda} \Delta x$$

- **Conditions for Constructive Interference (Bright Fringes):**

$$\Delta x = n\lambda, \quad \Delta \phi = 2n\pi \quad (n = 0, 1, 2, \dots)$$

- **Conditions for Destructive Interference (Dark Fringes):**

$$\Delta x = (2n - 1)\frac{\lambda}{2}, \quad \Delta \phi = (2n - 1)\pi \quad (n = 1, 2, \dots)$$

- **Position of Fringes in YDSE:**

$$y_n = \frac{n\lambda D}{d} \quad (\text{Bright}), \quad y'_n = (2n - 1)\frac{\lambda D}{2d} \quad (\text{Dark})$$

- **Fringe Width:**

$$\beta = \frac{\lambda D}{d}$$

- **Angular Fringe Width:**

$$\theta = \frac{\beta}{D} = \frac{\lambda}{d}$$

- **Shift of Fringe Pattern (due to thin film of thickness t , index n):**

$$\Delta y = \frac{D}{d}(n - 1)t$$

Single Slit Diffraction

- Condition for Minima (Slit width a):

$$a \sin \theta = n\lambda \quad (n = 1, 2, 3, \dots)$$

- Condition for Secondary Maxima:

$$a \sin \theta = (2n + 1)\frac{\lambda}{2} \quad (n = 1, 2, 3, \dots)$$

- Angular Width of Central Maximum:

$$2\theta = \frac{2\lambda}{a}$$

- Linear Width of Central Maximum:

$$\beta_0 = \frac{2\lambda D}{a} = 2\beta$$

11 Dual Nature of Radiation and Matter

Photons & Photoelectric Effect

- **Energy of a Photon:**

$$E = h\nu = \frac{hc}{\lambda} \approx \frac{1240 \text{ eV} \cdot \text{nm}}{\lambda \text{ (in nm)}} = \frac{12400 \text{ eV} \cdot \text{\AA}}{\lambda \text{ (in \AA)}}$$

- **Momentum of a Photon:**

$$p = \frac{E}{c} = \frac{h}{\lambda}$$

- **Effective Mass of Photon:**

$$m = \frac{h\nu}{c^2} = \frac{h}{c\lambda} \quad (\text{Rest mass } m_0 = 0)$$

- **Work Function (Φ_0):**

$$\Phi_0 = h\nu_0 = \frac{hc}{\lambda_0}$$

- **Einstein's Photoelectric Equation:**

$$K_{\max} = h\nu - \Phi_0 = h(\nu - \nu_0) = hc \left(\frac{1}{\lambda} - \frac{1}{\lambda_0} \right)$$

- **Cut-off / Stopping Potential (V_0):**

$$K_{\max} = eV_0 \implies eV_0 = h\nu - \Phi_0 \implies V_0 = \left(\frac{h}{e} \right) \nu - \frac{\Phi_0}{e}$$

(Slope of V_0 vs ν graph equals $\frac{h}{e}$, universal constant)

de Broglie Matter Waves

- de Broglie Wavelength of Particle (m, v):

$$\lambda = \frac{h}{p} = \frac{h}{mv}$$

- In Terms of Kinetic Energy K :

$$\lambda = \frac{h}{\sqrt{2mK}}$$

- For a Charged Particle Accelerated by Potential V :

$$\lambda = \frac{h}{\sqrt{2mqV}}$$

- For an Electron (m_e, e):

$$\lambda_e = \frac{1.227}{\sqrt{V}} \text{ nm} = \frac{12.27}{\sqrt{V}} \text{ \AA}$$

- For a Proton:

$$\lambda_p = \frac{0.286}{\sqrt{V}} \text{ \AA}$$

- For an Alpha Particle:

$$\lambda_\alpha = \frac{0.101}{\sqrt{V}} \text{ \AA}$$

- For Thermal Neutrons at Temperature T :

$$\lambda = \frac{h}{\sqrt{3mk_B T}}$$

12 Atoms

Rutherford Model & Bohr's Postulates

- Distance of Closest Approach (Impact Parameter $b = 0$):

$$r_0 = \frac{1}{4\pi\epsilon_0} \frac{2Ze^2}{K} = \frac{1}{4\pi\epsilon_0} \frac{4Ze^2}{mv^2}$$

- Bohr's Quantization Postulate:

$$L = mvr = \frac{nh}{2\pi} \quad (n = 1, 2, 3, \dots)$$

- Bohr's Frequency Condition:

$$h\nu = E_2 - E_1 \implies \frac{1}{\lambda} = \frac{E_2 - E_1}{hc}$$

Hydrogen-like Atom (Z) Formulas

- **Radius of n -th Orbit:**

$$r_n = \frac{n^2 h^2 \epsilon_0}{\pi m Z e^2} = (0.529 \text{ \AA}) \frac{n^2}{Z} \propto \frac{n^2}{Z}$$

- **Speed in n -th Orbit:**

$$v_n = \frac{Z e^2}{2 n h \epsilon_0} = \left(\frac{c}{137} \right) \frac{Z}{n} \propto \frac{Z}{n}$$

- **Kinetic Energy:**

$$K_n = \frac{1}{2} m v_n^2 = \frac{1}{4 \pi \epsilon_0} \frac{Z e^2}{2 r_n} = \frac{13.6 Z^2}{n^2} \text{ eV}$$

- **Potential Energy:**

$$U_n = -\frac{1}{4 \pi \epsilon_0} \frac{Z e^2}{r_n} = -2 K_n = -\frac{27.2 Z^2}{n^2} \text{ eV}$$

- **Total Mechanical Energy:**

$$E_n = K_n + U_n = -K_n = \frac{U_n}{2} = -\frac{13.6 Z^2}{n^2} \text{ eV}$$

- **Rydberg Formula (Wave Number $\bar{\nu}$):**

$$\bar{\nu} = \frac{1}{\lambda} = R Z^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right) \quad (R \approx 1.097 \times 10^7 \text{ m}^{-1})$$

- **Spectral Series of Hydrogen ($Z = 1$):**

- **Lyman Series (Ultraviolet):** $n_1 = 1, n_2 = 2, 3, 4, \dots$
- **Balmer Series (Visible):** $n_1 = 2, n_2 = 3, 4, 5, \dots$
- **Paschen Series (Infrared):** $n_1 = 3, n_2 = 4, 5, 6, \dots$
- **Brackett Series (Infrared):** $n_1 = 4, n_2 = 5, 6, 7, \dots$
- **Pfund Series (Far Infrared):** $n_1 = 5, n_2 = 6, 7, 8, \dots$

13 Nuclei

Nuclear Properties & Mass-Energy Equivalence

- **Nuclear Radius:**

$$R = R_0 A^{1/3} \quad (R_0 \approx 1.2 \times 10^{-15} \text{ m} = 1.2 \text{ fm})$$

- **Nuclear Density:**

$$\rho = \frac{\text{Mass}}{\text{Volume}} \approx 2.3 \times 10^{17} \text{ kg/m}^3 \quad (\text{Independent of Mass Number } A)$$

- **Atomic Mass Unit (amu or u):**

$$1 \text{ u} = 1.6605 \times 10^{-27} \text{ kg} = 931.5 \text{ MeV}$$

- **Mass Defect (Δm):**

$$\Delta m = [Zm_p + (A - Z)m_n] - M_{\text{nucleus}}$$

- **Nuclear Binding Energy:**

$$E_b = \Delta m \cdot c^2 = (\Delta m \text{ in u}) \times 931.5 \text{ MeV}$$

- **Binding Energy per Nucleon:**

$$\bar{E}_b = \frac{E_b}{A} \quad (\text{Peaks at } \sim 8.75 \text{ MeV for } {}^{56}\text{Fe})$$

- **Nuclear Fission / Fusion Q -Value:**

$$Q = [m_{\text{reactants}} - m_{\text{products}}] \times 931.5 \text{ MeV}$$

14 Semiconductor Electronics

Semiconductor Physics and Diodes

- **Charge Carrier Density (Mass-Action Law):**

$$n_e \cdot n_h = n_i^2$$

- Intrinsic semiconductor: $n_e = n_h = n_i$
- n-type semiconductor: $n_e \approx N_d \gg n_h$
- p-type semiconductor: $n_h \approx N_a \gg n_e$

- **Total Electric Current:**

$$I = I_e + I_h = eA(n_e v_e + n_h v_h) = eAE(n_e \mu_e + n_h \mu_h)$$

- **Conductivity of Semiconductor:**

$$\sigma = e(n_e \mu_e + n_h \mu_h)$$

- **Dynamic Resistance of p-n Junction Diode:**

$$r_d = \frac{\Delta V}{\Delta I}$$

- **Half-Wave Rectifier:**

$$f_{\text{out}} = f_{\text{in}}, \quad V_{\text{dc}} = \frac{V_0}{\pi} \approx 0.318V_0, \quad \eta_{\text{max}} = 40.6\%$$

- **Full-Wave Rectifier:**

$$f_{\text{out}} = 2f_{\text{in}}, \quad V_{\text{dc}} = \frac{2V_0}{\pi} \approx 0.636V_0, \quad \eta_{\text{max}} = 81.2\%$$